```
!pip install pgmpy
```

```
Collecting pgmpy
  Downloading pgmpy-1.1.0-py3-none-any.whl.metadata (13 kB)
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Installing collected packages: scikit-base, pgmpy
Successfully installed pgmpy-1.1.0 scikit-base-0.13.1
```

```
# Install pgmpy if not already installed
# pip install pgmpy

from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.estimators import MaximumLikelihoodEstimator
from pgmpy.inference import VariableElimination
import pandas as pd

data = pd.DataFrame({
    'Rain': ['No', 'No', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'No'],
    'TrafficJam': ['Yes', 'No', 'Yes', 'No', 'Yes', 'Yes', 'No', 'No'],
    'ArriveLate': ['Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'Yes', 'No']
})

print("Dataset:\n", data, "\n")

model = DiscreteBayesianNetwork([
    ('Rain', 'TrafficJam'),
    ('TrafficJam', 'ArriveLate')
])

# ----------------------------------------
# Step 3: Fit model using MLE
# ----------------------------------------
model.fit(data, estimator=MaximumLikelihoodEstimator)

# ----------------------------------------
# Step 4: Print CPDs (Conditional Probabilities)
```

```
# ----------------------------------------
print("Conditional Probability Distributions (CPDs):\n")
for cpd in model.get_cpds():
    print(cpd)
    print()

# ----------------------------------------
# Step 5: Perform inference
# ----------------------------------------
inference = VariableElimination(model)

# Query: P(ArriveLate | Rain = Yes)
query_result = inference.query(
    variables=['ArriveLate'],
    evidence={'Rain': 'Yes'}
)

# ----------------------------------------
# Step 6: Display result
# ----------------------------------------
print("Inference Result: P(ArriveLate | Rain = Yes)\n")
print(query_result)

# ----------------------------------------
# Optional: Query more cases
# ----------------------------------------
print("\nAdditional Queries:\n")

# P(ArriveLate | Rain = No)
result2 = inference.query(
    variables=['ArriveLate'],
    evidence={'Rain': 'No'}
)
print("P(ArriveLate | Rain = No):")
print(result2)

# P(TrafficJam | Rain = Yes)
result3 = inference.query(
    variables=['TrafficJam'],
    evidence={'Rain': 'Yes'}
)
print("\nP(TrafficJam | Rain = Yes):")
print(result3)
```

```
5  Yes        Yes        Yes
6  Yes         No        Yes
7   No         No         No

Conditional Probability Distributions (CPDs):

+-----------+-----+
| Rain(No)  | 0.5 |
+-----------+-----+
| Rain(Yes) | 0.5 |
+-----------+-----+

+-----------------+----------+-----------+
| Rain            | Rain(No) | Rain(Yes) |
+-----------------+----------+-----------+
| TrafficJam(No)  | 0.5      | 0.5       |
+-----------------+----------+-----------+
| TrafficJam(Yes) | 0.5      | 0.5       |
+-----------------+----------+-----------+

+-----------------+----------------+-----------------+
| TrafficJam      | TrafficJam(No) | TrafficJam(Yes) |
+-----------------+----------------+-----------------+
| ArriveLate(No)  | 0.75           | 0.25            |
+-----------------+----------------+-----------------+
| ArriveLate(Yes) | 0.25           | 0.75            |
+-----------------+----------------+-----------------+
```

```
+-----------------+-------------------+

Additional Queries:

P(ArriveLate | Rain = No):
+-----------------+-------------------+
| ArriveLate      |   phi(ArriveLate) |
+=================+===================+
| ArriveLate(No)  |            0.5000 |
+-----------------+-------------------+
| ArriveLate(Yes) |            0.5000 |
+-----------------+-------------------+

P(TrafficJam | Rain = Yes):
+-----------------+-------------------+
| TrafficJam      |   phi(TrafficJam) |
+=================+===================+
| TrafficJam(No)  |            0.5000 |
+-----------------+-------------------+
| TrafficJam(Yes) |            0.5000 |
+-----------------+-------------------+
```